Relief Forge

The basics. One picture in, one carvable relief out.

CNC · Fibre laser · 3D print · Moulds · Coins & medals

Quick start guide · October 2026

snazzygaz.github.io/DepthNormal

Runs in your web browser · nothing is uploaded anywhere

THE GOAL

What this thing does

Relief Forge turns a **picture** into a **height map**: a grey image where **white = high** and **black = low**. CNC routers, fibre lasers and 3D printers read that grey as depth, so the picture comes out as a carved, raised object, like a coin, a medal or a stone panel.

Two AI models look at your picture. One guesses **how far away** everything is (depth). The other guesses **which way each bit of surface faces** (normals). The Forge blends the two into a clean relief and gives you sliders to shape it.

Everything happens on your own computer. Your pictures never leave it. The AI models download once (the first time) and are then kept by your browser.

What you need

- **A computer with Google Chrome** (or Edge). A decent graphics card makes it fast. Without one it still works, just slowly.
- **About 1 GB of free space** for the AI models, and an internet connection for the first run.
- **A good picture.** This matters more than any setting:
  - High resolution, sharp, in focus.
  - The subject **cut out** from its background: a transparent background or one plain colour (all orange, all white…).
  - Even, soft lighting. No hard shadows.
  - AI image editing can help clean up a picture first, e.g. remove the background or tidy it up.

HOW IT FLOWS

The four steps

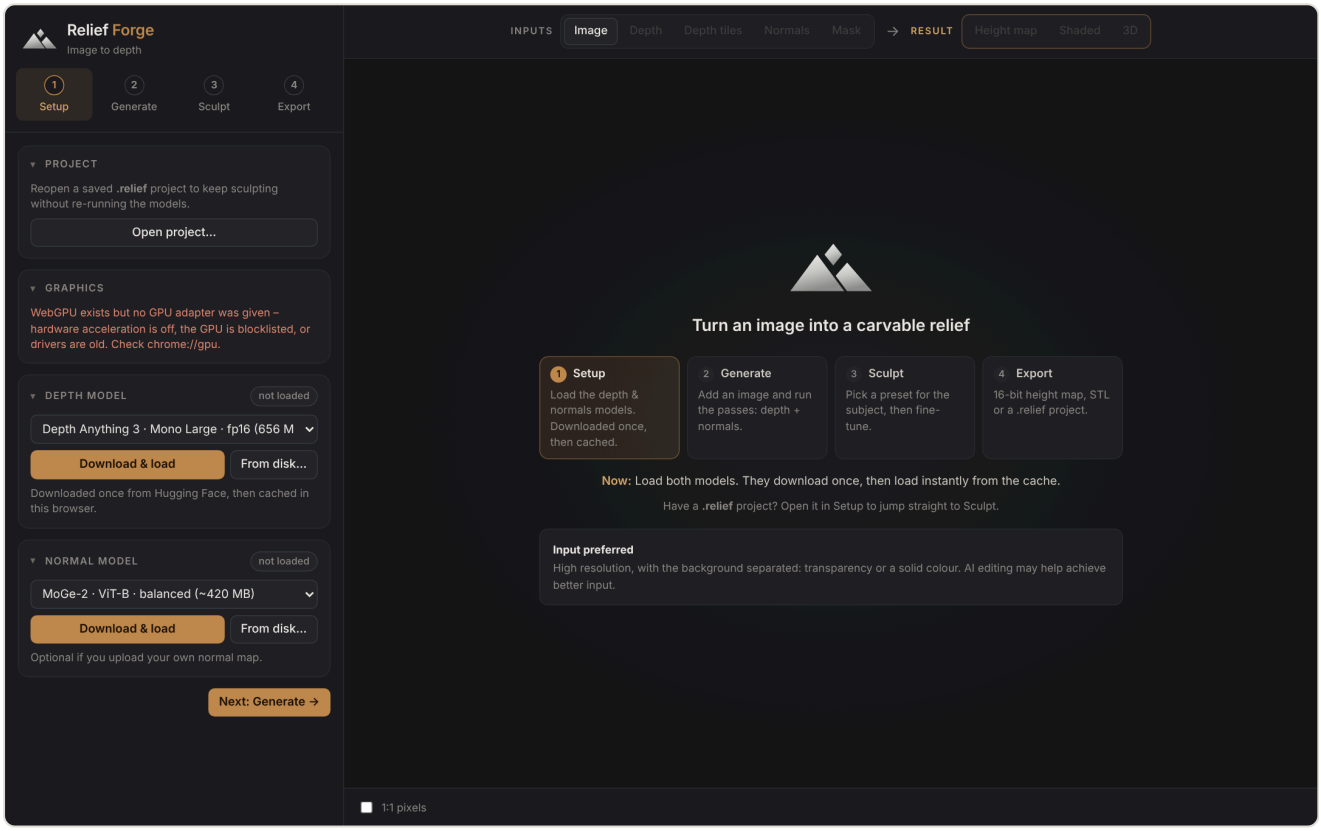
1
Setup
Load the two AI models. Only slow the first time.

2
Generate
Add your picture, press **Run passes**.

3
Sculpt
Pick a preset, nudge sliders, watch it update.

4
Export
Save the height map, an STL or the project.

The buttons for these four steps are at the top left. Do them in order. The big area on the right tells you what to do next (the **“Now:”** line).



The start screen. Left: the controls for the step you’re on. Right: what to do next.

STEP 1

Setup

Project

Already made something and saved it? **Open project…** loads it back and jumps straight to Sculpt. No AI needed.

Graphics

Tells you if your graphics card can be used. If it can't, everything still works on the processor, just slower.

Depth model

The AI that guesses distance. Press **Download & load**. Leave it on the first choice (Depth Anything 3) unless you know why you want another. Depth Anything 2 is also offered; it is **non-commercial only**.

Normal model

The AI that guesses surface direction. It gives all the crisp detail. Press **Download & load**. ViT-B is the sensible default; ViT-L is bigger and slower.

STEP 2

Generate

Source image

Drop your picture in, or click to choose it.

Working size

How big (in pixels) the relief is made. 2048 is a good default. Bigger = more detail, but slower.

Depth tiles / Normal tiles

The picture is cut into pieces, each piece is looked at closely, then they are stitched back together. **3 across** gives much more detail. Turn off for speed.

Normal detail (tokens)

How finely the normal AI looks at each piece. 2500 is the default. Higher is sharper but heavier on your graphics card.

Run passes

Press it and wait. When it says **Done**, go to Sculpt.

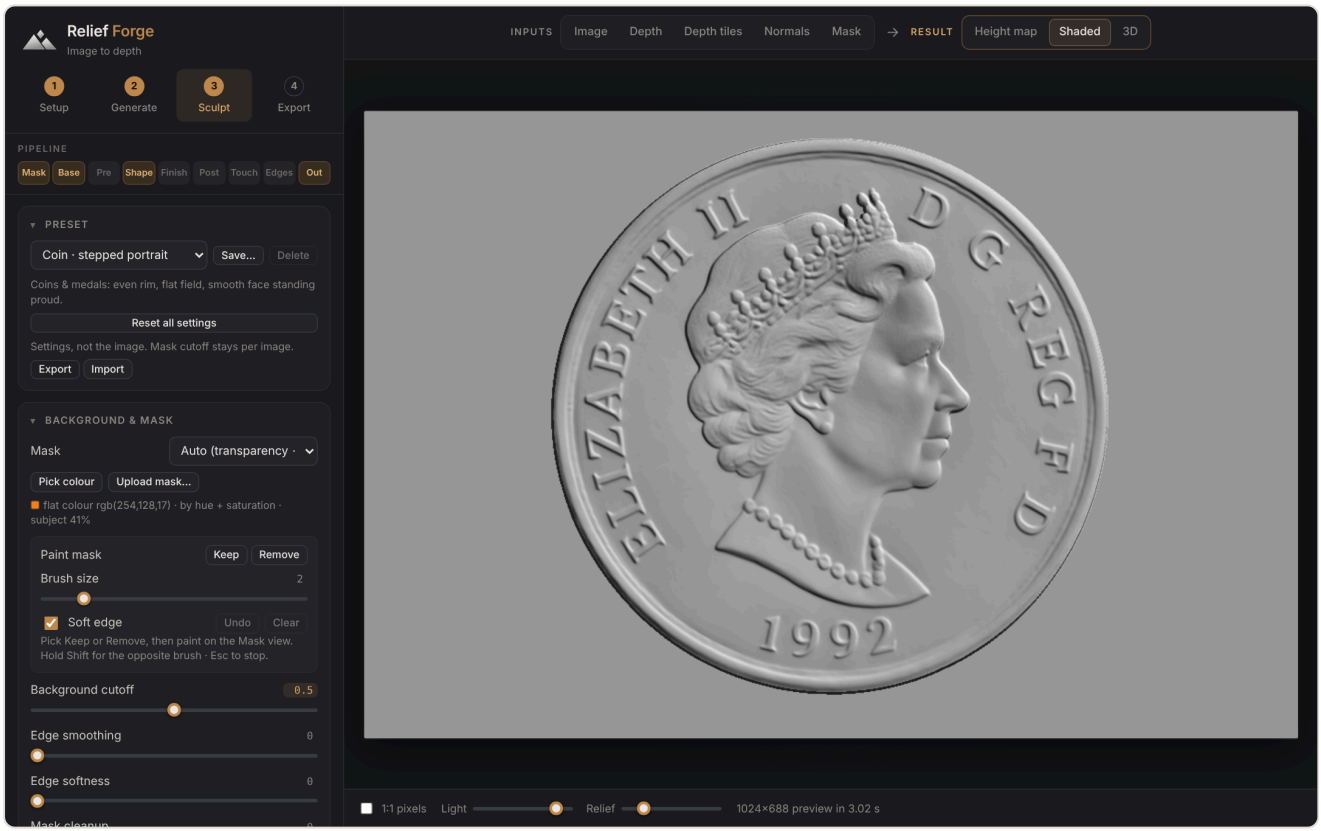
If it crashes with "device lost": close other heavy programs, lower Normal detail, reload the page and try again.

LOOKING AT THINGS

The view tabs (top of the screen)

Inputs (left group) are the raw materials. **Result** (right group, in bronze) is what you'll actually get.

TAB	WHAT IT SHOWS
Image	Your picture.
Depth / Depth tiles	What the depth AI saw. White = near.
Normals	What the normal AI saw, as colours.
Mask	Red = background that gets thrown away.
Height map	**The actual output.** White = high.
Shaded	The height map lit from the side, so it looks like the real object. Easiest to judge.
3D	A real 3D model you can spin, in different materials.



The Sculpt step with the "Coin · stepped portrait" preset, Shaded view.

STEP 3

Sculpt, section by section

The panels run top to bottom in the order the work is done. Click a panel's title to fold it away. The **Pipeline** strip at the top lights up the stages you've changed; click one to jump to it.

Which preset?

YOUR SUBJECT	PRESET
Not sure / anything	**NHeavyDat** (Starting points, near the top). A strong all-round start.
Statue, figure, carving	**Statue · shallow** for a flatter relief, **Statue · deep** for more volume.
Coin or medal with a face	**Coin · stepped portrait**, or **Coin · crisp, keeps steps (OG2)**.
Busy challenge coin	**Coin · challenge / busy art**
Badge, crest, emblem	**Emblem · stepped crest**
Start from scratch	**Neutral · base only** (the very top), then build up yourself.

Preset

Start here. Pick the one that matches your subject (see the table above). **Starting points** sit at the top of the list: **Neutral** (everything off) and **NHeavyDat** (good all-rounder). A note under the box says what it's for. **Reset all settings** puts everything back to plain. **Save…** keeps your own settings; **Export / Import** shares them as a file.

Background & mask

Decides what is subject and what is background. **Auto** usually gets it right. If not, press **Pick colour** and click the background. Paint with **Keep / Remove** to fix small mistakes. The background becomes flat black.

Base & pre layers

What the relief is built from. **Auto** is fine. "Normals" gives smoother, cleaner shapes. Pre layers let you mix in other maps; leave them alone at first.

Shape

The big moves: how much fine detail to add, how deep or flat the whole thing is, and levelling so it isn't tilted. **Level field** and **Auto-level tilt** make backgrounds and coin fields flat.

Finish

Clean-up and polish: **Denoise**, the **Detail EQ** (fine, medium, broad), **Crease deepen** for crisp lines, gamma and a tone curve.

Post layers

Blends another map over the finished result, e.g. a little texture. Optional.

Touch-up brush

Paint directly on the result: **Raise**, **Lower**, **Smooth**, **Erase**. Has its own Undo, and an on/off switch to compare.

Edges

How the subject meets the background. **Edge roll-off** makes it slope down softly instead of a cliff. **Lift subject** raises it off the floor like a slab (coins, badges).

Output

Stretch output uses the full black-to-white range. **Invert** turns it inside out (for a mould). **Mirror for mould** flips it so a cast reads the right way round.

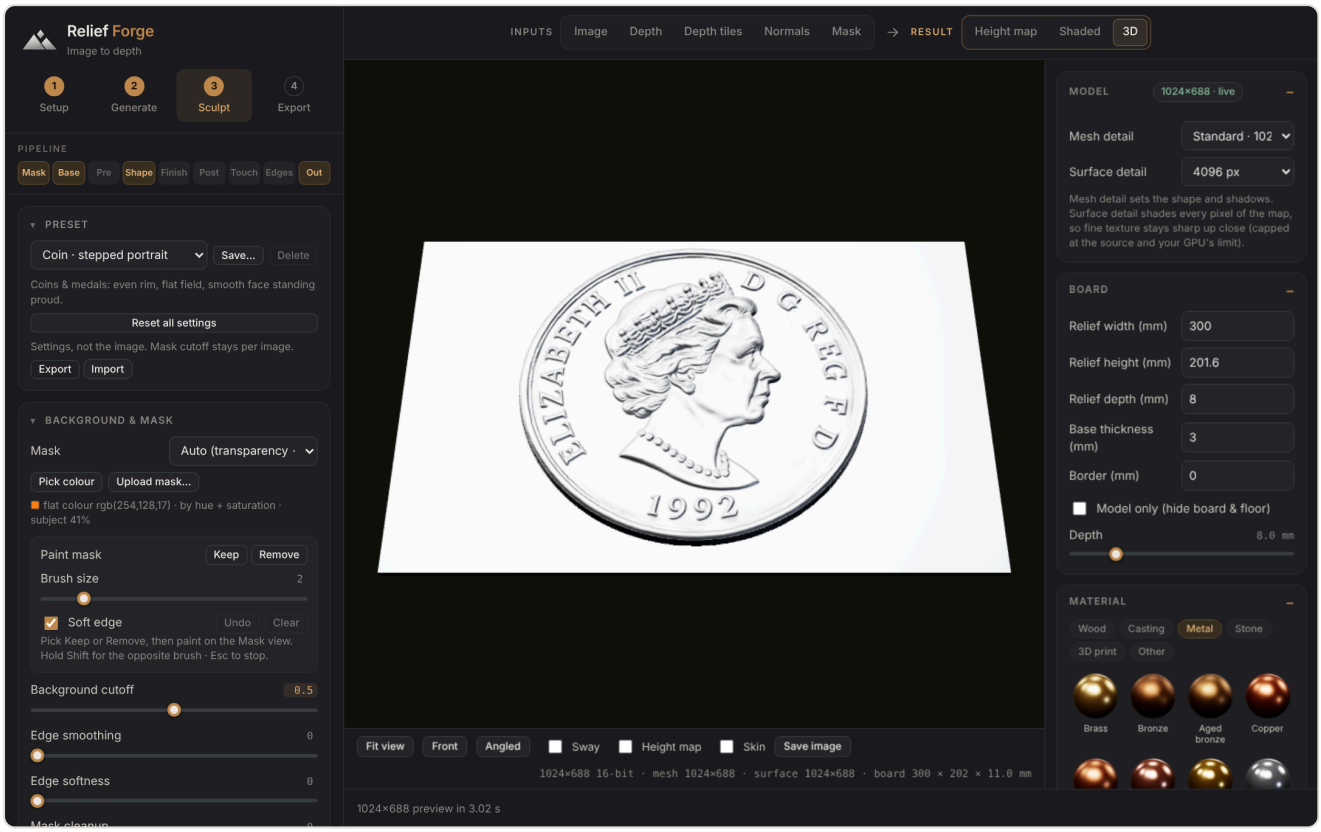
Advanced

Fine controls most people never need.

Live updates while you drag. **Fast preview** works at half size while you tweak; exports are always full size.

CHECKING IT

The 3D view



The 3D tab, here in Steel. The model updates live as you change sliders.

Spin it

Drag to turn, scroll to zoom. **Fit view**, **Front** and **Angled** reset the camera.

Material

Click a ball to see it in steel, bronze, wood, stone, resin and more.

Board

Size and depth of the panel in millimetres, plus base thickness and border.

Light

Pick a lighting preset or move the light around.

Height map / Skin (bottom bar)

Height map shows the raw grey on the model. **Skin** wraps your original picture over it, to check things line up. Untick both to go back to the material.

Save image

Saves a picture of the current view.

STEP 4

Export

File name

The name used for everything you save. Leave blank for an automatic one.

Height map

Download 16-bit PNG: the main thing. Use this for CNC and lasers (smooth steps). 8-bit is only for programs that can't read 16-bit.

3D model · STL

A solid 3D file for printers and CAM. Set panel width, relief depth and backing. It can also build a **silicone pour box** around it and **split it** to fit your print bed.

Normal map

The colour surface map, for 3D artists.

Project

Save project keeps everything (picture, AI results, settings, brushwork) in one .relief file. Open it later in Setup and carry on without running the AI again.

WHEN THINGS GO WRONG

Quick fixes

PROBLEM	TRY THIS
Background is part of the relief	Background & mask: set Mask to Auto or Flat colour, or paint Remove over it.
Whole thing looks tilted	Shape: raise Auto-level tilt and Level field.
Grainy or noisy surface	Finish: raise Denoise, lower Fine detail. Use a sharper picture.
Too flat, no depth	Use the Statue · deep preset, or lower Compress depth.
Hard cliff around the edge	Edges: raise Edge roll-off.
“Device lost” error	Close heavy programs, lower Normal detail, reload the page.
Very slow	Use a smaller Working size and fewer tiles, and keep Fast preview on.
Messed it all up	Preset: **Reset all settings**.

THE SMALL PRINT

Licences & credits

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Relief Forge uses these third-party parts, each under its own licence:

COMPONENT	BY	LICENCE
Depth Anything 3 · Mono Large	ByteDance Seed (Depth Anything team)	Apache 2.0
DA3 Mono Large ONNX conversion	Jens Duttke	Apache 2.0
Depth Anything V2 · Large / Base `optional`	Depth Anything team; ONNX by onnx-community	CC BY-NC 4.0 · **non-commercial use only**
MoGe-2 normal models	Microsoft	MIT
DINOv2 (inside MoGe-2)	Meta AI	Apache 2.0
ONNX Runtime Web	Microsoft	MIT
three.js (3D view)	three.js authors	MIT
Manrope, IBM Plex Mono (fonts)	Mikhail Sharanda; IBM	SIL Open Font Licence 1.1

Commercial work: if you sell what you make, use Depth Anything 3 for depth. Depth Anything V2 Large and Base are for non-commercial use only.

The AI models are downloaded from Hugging Face the first time you use them and are not part of the Relief Forge app itself. Full licence texts are on each project's page. This summary is not legal advice.